

Q1: Define Linear Regression concept for best fit line and compare with Logistic Regression?

Ans:- Linear Regression:-

- It is used to predict continuous values.
- It finds the best fit line through data points using the formula:

$$y = mx + c$$
- Uses Mean Squared Error (MSE) as loss function.

Logistic Regression:-

- It is used to predict categorical values
- Instead of a straight line, it gives an S-shaped (sigmoid) curve.
- Uses Log Loss instead of MSE.

Comparison:-

<u>Feature</u>	<u>Linear Regression</u>	<u>Logistic Regression</u>
Output	Continuous value	0 or 1
Graph	Straight line	S-curve
use	Prediction of numbers	Classification
Loss function	MSE	Log Loss

Q28 Find out min-max Normalization and Z-Score Standardization for marks: 50, 60, 70, 80, 90 ?

Ans: Given data

Marks = 50, 60, 70, 80, 90

Min-Max Normalization

Formula :-
$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Here :

$$X_{\min} = 50$$

$$X_{\max} = 90$$

Now calculate for each mark :

marks(x)	calculation	Normalization value.
50	$(50-50)/(90-50)$	0
60	$(60-50)/(90-50)$	0.25
70	$(70-50)/(90-50)$	0.50
80	$(80-50)/(90-50)$	0.75
90	$(90-50)/(90-50)$	1

Min - Max Normalized values :-

0, 0.25, 0.5, 0.75, 1

Z-Score Standardization

Formula:

$$Z = \frac{X - \text{Mean}}{\text{Standard Deviation}}$$

Step 1:- Find Mean (μ)

$$\text{Mean} = \frac{50 + 60 + 70 + 80 + 90}{5} = 70$$

Step 2:- Find Standard Deviation (σ)

$$\begin{aligned} \sigma &= \sqrt{\frac{(50-70)^2 + (60-70)^2 + (70-70)^2 + (80-70)^2 + (90-70)^2}{5}} \\ &= \sqrt{\frac{400 + 100 + 0 + 100 + 400}{5}} \\ &= \sqrt{\frac{1000}{5}} \Rightarrow \sqrt{200} = 14.14 \end{aligned}$$

Step 3:- calculate Z for each mark

Mark (X)	calculation	Z-Score
50	$(50-70)/14.14$	-1.41
60	$(60-70)/14.14$	-0.71
70	$(70-70)/14.14$	0
80	$(80-70)/14.14$	+0.71
90	$(90-70)/14.14$	+1.41

Z-Score values

-1.41, -0.71, 0, +0.71, +1.41

Q3:- How to use Gradient Descent Algorithm in Linear Regression with ~~derivation~~ derivation?

Ans:-

- In Linear Regression, we try to find a line:

$$y = mx + c$$

m = slope,

c = intercept

- Cost Function (Error Function)

we measure how wrong our line is using the Mean Squared Error (MSE):

$$J(m, c) = \frac{1}{2n} \sum_{i=1}^n (y_i - (mx_i + c))^2$$

where:

y_i = actual value

$mx_i + c$ = predicted value

n = total number of data points

- Goal of Gradient Descent

we want to minimize $J(m, c)$ by updating m and c step-by-step in the direction of minimum error.

- Derivation

Take the partial derivatives of J with respect to m and c .

- (i) for slope m :

$$\frac{\partial J}{\partial m} = -\frac{1}{n} \sum (y_i - (mx_i + c)) \cdot x_i$$

- (ii) for intercept c :

$$\frac{\partial J}{\partial c} = -\frac{1}{n} \sum (y_i - (mx_i + c))$$

Q4:- What is the Association Rule? Explain Support, confidence and Lift?

Ans:-

Association Rule

• Association Rule learning is a technique used in data mining to find relationship or pattern among items in large datasets.

Example:-

In a supermarket if many customers buy bread, they also buy butter.

→ Rule: Bread → Butter

Support:-

- It shows how often items appear together in all transactions.
- It helps in find how frequent a rule is in the dataset.

$$\text{Support}(A \rightarrow B) = \frac{\text{Transactions containing both A \& B}}{\text{Total transaction}}$$

Confidence:-

- It shows how often B is bought when A is bought.
- It measures the strength of the rule:

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \rightarrow B)}{\text{Support}(A)}$$

Lift:-

- Lift measures how much more likely items bought together than by random chance.

$$\text{Lift}(A \rightarrow B) = \frac{\text{Confidence}(A \rightarrow B)}{\text{Support}(B)}$$

Q5: Draw a 4x5 table Day D1, D2, D3, D4, D5.

	outlook	wind	Play Tennis
D1	Sunny	Strong	Yes
D2	Sunny	strong	No
D3	Rainy	strong	Yes
D4	Rainy	weak	No
D5	Rainy	weak	No

Ans:

1. Total entropy of Play Tennis

Data: yes = 2, No = 3, total = 5

Probabilities:

$$P(\text{yes}) = 2/5 = 0.4$$

$$P(\text{No}) = 3/5 = 0.6$$

Entropy formula: $H = - \sum p \log_2 p$

calculation:

$$\bullet -0.4 \log_2 0.4 = 0.528771$$

$$\bullet -0.6 \log_2 0.6 = 0.442179$$

• Total:

$$H = 0.528771 + 0.442179 = 0.970951$$

$$\boxed{\text{Entropy (Play Tennis)} \approx 0.971}$$

2. Information Gain (IG) for outlook

• outlook values:

• Sunny $\rightarrow$ D1, D2 $\rightarrow$ Play Tennis: [yes/No]

$\rightarrow$ counts (yes = 1, No = 1)

• Rainy $\rightarrow$ D3, D4, D5 $\rightarrow$ Play Tennis: [yes, No]

$\rightarrow$ counts (yes = 1, No = 2)

• Entropy per branch :

- Entropy (Sunny) :

→ Probabilities 0.5 & 0.5 → entropy = 1.000

- Entropy (Rainy) :

→ $P(\text{yes}) = 1/3 \approx 0.333$, $P(\text{No}) = 2/3 \approx 0.667$

→ Entropy = $-\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \approx 0.918296$

• Weighted average entropy (outlook) :

• Sunny weight = $2/5 = 0.4$ → $0.4 \times 1.000 = 0.400000$

• Rainy weight = $3/5 = 0.6$ → $0.6 \times 0.918296 = 0.550978$

• weighted entropy = $0.400000 + 0.550978 = 0.950978$

• Information Gain :

$IG(\text{outlook}) = \text{Entropy}(\text{total}) - \text{weighted entropy}$

= $0.970951 - 0.950978$

= 0.019973

$\boxed{IG(\text{outlook}) \approx 0.020}$

• Information Gain (IG) for wind

• wind values :

• Strong → D1, D2, D3 → Play Tennis : [yes, No, yes]

→ counts (yes = 2, No = 1)

• weak → D4, D5 → Play Tennis : [No, No] → count (yes = 0, No = 2)

• Entropy (~~Strong~~) per branch :

• Entropy (Strong) :

→ $P(\text{yes}) = 2/3 \approx 0.6667$, $P(\text{No}) = 1/3 = 0.3333$

→ entropy = 0.918296

• Entropy (weak) :

→ $P(\text{yes}) = 0$, $P(\text{No}) = 1$ → entropy = 0.000

• weighted average entropy (wind) %

• Strong weight = $3/5 = 0.6 \rightarrow 0.6 \times 0.918296 = 0.550978$

• weak weight = $2/5 = 0.4 \rightarrow 0.4 \times 0 = 0.0000$

• weighted entropy = $0.550978 + 0 = 0.550978$

• Information Gain (wind) %

$$I_1(\text{wind}) \approx 0.420$$

3. Choose the root node %

• Compare $I_1(s)$:

• $I_1(\text{outlook}) \approx 0.020$

• $I_1(\text{wind}) \approx 0.420$

• Highest $I_1 = \text{wind}$

• Root node = wind

4. First level of the decision tree :-

Root : wind

• Strong $\rightarrow$ Subset (D_1, D_2, D_3) $\rightarrow$ mixed (Yes/No) $\rightarrow$ needs further split

• weak $\rightarrow$ (D_4, D_5) $\rightarrow$ Play Tennis = No

